ROM [Read Only Memory]

→ It is non Voletile memory.

→ So, when Power is OFF, content of memory will stay stored.

Rom structure.

= Decoder + Programmble OR gates.

= AND gates (fixed) + Programmble OR gates.

ROM size.

$= 2^x \times y$

where, x = Number of I/p

y = Number of O/p.

→ E.g. If I/p = 8, O/p = 4 then what is the size of Rom

$x = 8$, $y = 4$

Size $= 2^x \times y$

$= 2^8 \times 4 = 1024$ bits

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$= (1024/8)$ bytes

$= 128$ bytes.

$= (1024/8)$ bytes

$= 128$ bytes.

ROM Classifications

PROM
- Programmable ROM
- Initially it is empty
- Then we can program it once
- Data can not be erased here.

EPROM
- Erasable PROM
- By UV rays we can erase data.
- After that we can reprogram this memory.

EEPROM
- Electrically EPROM
- we can erase data Electrically many times.
- Efficiency of memory will decay with respect to erase.
- Erase happens bit by bit.

Flash memory
- Speed of erase is faster then EEPROM.
- Data erase is done block by block.

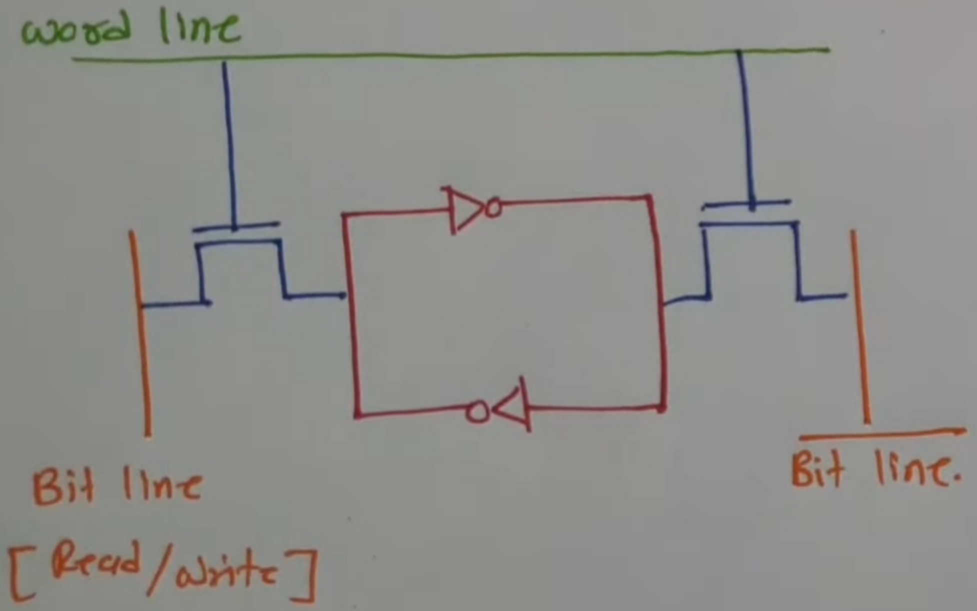
Mask Rom
- It is PROM only.
- It is programmed by chip manufacturer.
- this ROM is masked OFF during photolithography.

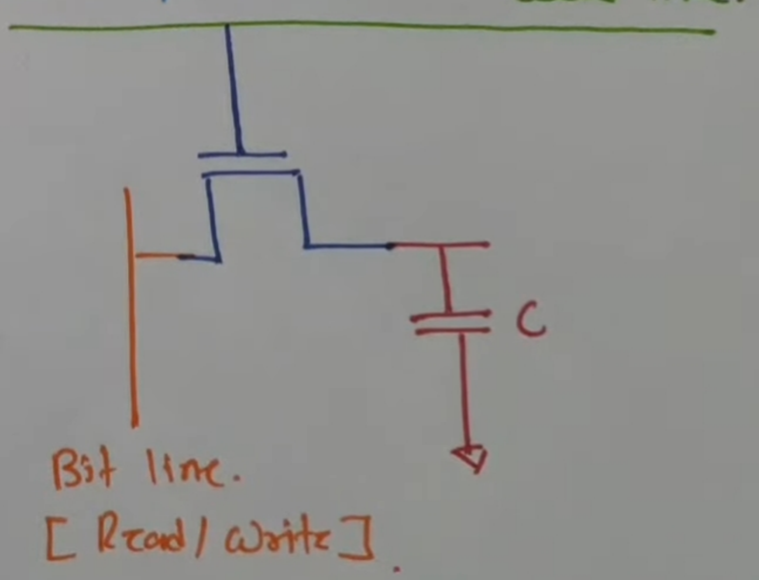
RAM [Random Access Memory]

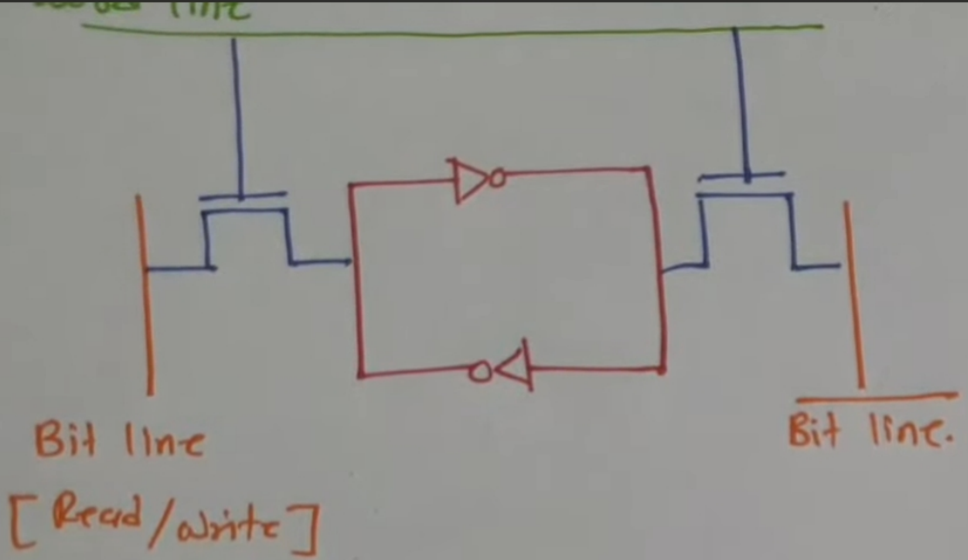
→ It is Voletile memory.

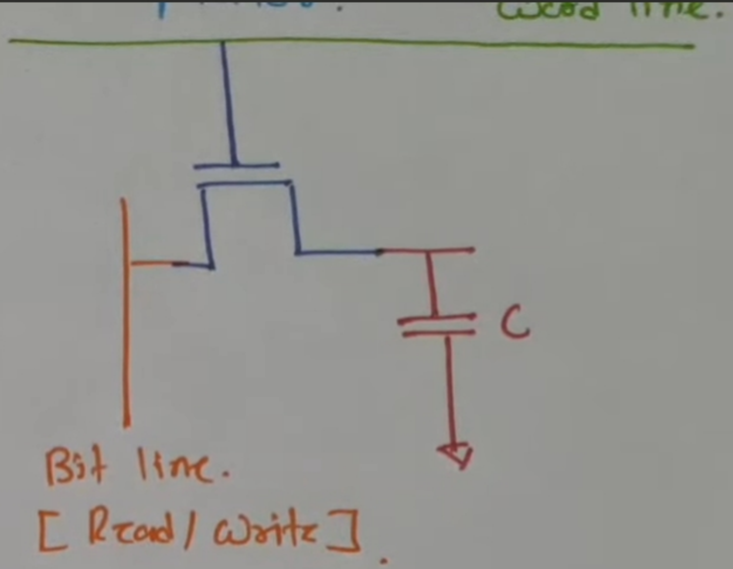
→ So, If power is OFF, then data of RAM will get erased.

SRAM	DRAM
→ Static RAM	→ Dynamic RAM
→ It is made up of Flip Flop.	→ It is made up of Transistor and capacitor.









→ Faster
→ No need of periodic refresh.
→ high cost
→ structure density is higher
→ It needs more power
→ It generates more heat
→ It is used in CPU cache

→ Slower.
→ It needs periodic refresh of capacitor.
→ low cost.
→ Structure density is lower.
→ It needs ~~more~~ less power.
→ It generates less heat.
→ It is used in main memory of computer.